

Use Secret Manager

(LAB-M13-01)

Overview

In this lab, you will be deploying web server with database.

You are securing the database credentials by storing in secret manager and accessed it programmatically.

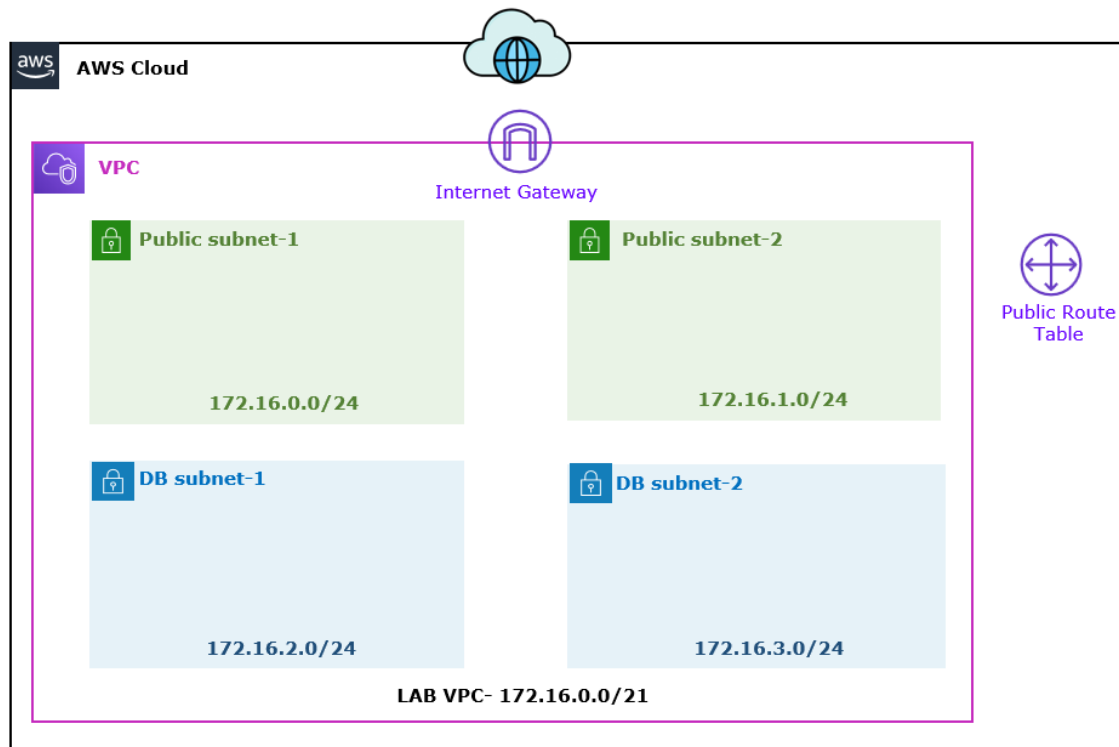
Objectives

After completing this lab, you will be able to:

- Create Secret manager.
- Save database secrets in Secret manager.
- Access database secrets from Secret manager.

Task 1: Create VPC

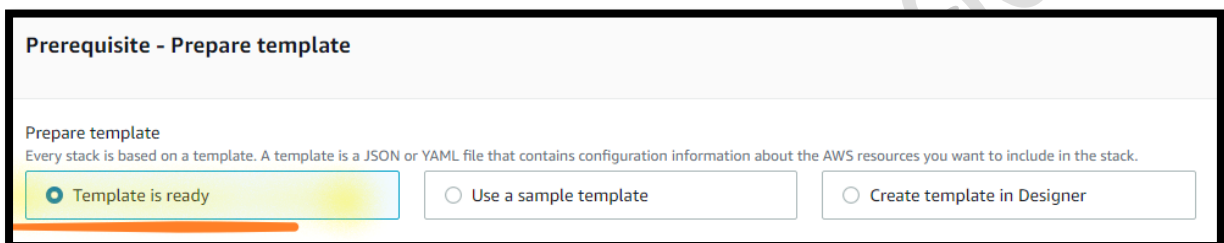
In this task you can create VPC with subnets.



Step 1: Deploy a VPC using CloudFormation

AWS CloudFormation model and provision AWS resources in your cloud environment in automated way.

1. In the **AWS Management Console**, on the **Services** menu, click **CloudFormation**.
2. Dropdown and select the **US East (N. Virginia)** region.
3. Click **Create stack**:
 - a. In the **Prerequisite - Prepare template** section:
 - i. **Prepare template**: Select **Template is ready**.



Prerequisite - Prepare template

Prepare template
Every stack is based on a template. A template is a JSON or YAML file that contains configuration information about the AWS resources you want to include in the stack.

☒ Template is ready ☐ Use a sample template ☐ Create template in Designer

- b. In the **Specify template** section:
 - i. **Template source**: Select **Upload a template file**.
 - ii. **Upload a template file**: Click on **choose file** then select the **LAB-VPC.yaml** file.

Note: **LAB-VPC.yaml** template is provided with the lab manual.

- iii. Click **Next**.
 - c. In the **Stack name** section:
 - i. **Stack name**: Write **LABVPC**.
 - d. In the **Parameters** section:
 - i. **Environment name**: Write **LAB VPC**.

Note: Leave other values as default.

- ii. Click **Next**.

e. In the **Configure stack options** page:

i. Select **Add Tag**:

i. **Key**: Write **Name**.

ii. **Value**: Write **LAB VPC**.

Note: Leave all the details as default.

ii. Click **Next**.

f. In the **Review LABVPC** page:

i. Click **Create stack**.

ii. The stack status (*left site under Stack*) is showing as **CREATE_IN_PROGRESS**.

Note: Click on **Refresh** until status is turned into **CREATE_COMPLETE**.

Note: You need to wait till you can see the **CREATE_COMPLETE** status **under Stack** on your **right site**, as shown below.

Note: **Ignore** if you can see any resource **CREATE_IN_PROGRESS** under **Events**.

4. **From** the **LABVPC-01** Stack:

a. **Go to right**, click the **Outputs** tab.

b. A **CloudFormation stack** can provide **output information**, such as resources details. **You will see outputs**:

i. **PrivateSubnets**: The **Id of the DB Subnets** that was created.

ii. **PublicSubnets**: The **Id of the Public Subnets** that was created.

iii. **VPC**: The **Id of the VPC** that was created.

Task 2: Create Security Groups

In this task you can create security groups for web server, database instance and secret manager.

Step 1: Create WebApp Security Group

5. **From** the **VPC** console.
6. **Go to the left**, select **Security Groups**.
7. Select **Create security group**.
 - a. In the **Basic details** section:
 - i. **Security group name**: Write **WebApp-SG**.
 - ii. **Description**: Write **Security Group for WebApp Server**.
 - iii. **VPC**: Click on **Cross**.
 - a) Select **LAB VPC**.
 - b. In the **Inbound rule** section:
 - i. Click **Add Rule**.
 - a) **Type**: Dropdown and select **SSH**.
 - b) **Source**: Dropdown and select **Anywhere-IPv4**.
 - ii. Click **Add Rule**.
 - c) **Type**: Dropdown and select **HTTP**.
 - d) **Source**: Dropdown and select **Anywhere-IPv4**.
 - c. In the **Tags - optional** section:
 - i. Click **Add new tag**:
 - a) **Key**: Write **Name**.
 - b) **Value**: Write **WebApp-SG**.
 - d. Click **Create security group**.

Step 2: Create Database Security Group

8. **From** the **VPC** console.
9. **Go to the left**, select **Security Groups**.
10. Select **Create security group**.
 - a. In the **Basic details** section:
 - i. **Security group name**: Write **Database-SG**.
 - ii. **Description**: Write **Security Group for DB Server**.
 - iii. **VPC**: Click on **Cross**.
 - a) Select **LAB VPC**.
 - b. In the **Inbound rule** section:
 - i. Click **Add Rule**.
 - a) **Type**: Dropdown and select **MySQL/Aurora**.
 - b) **Source**: Dropdown and select **Anywhere-IPv4**.
 - c. In the **Tags - optional** section:
 - i. Click **Add new tag**:
 - a) **Key**: Write **Name**.
 - b) **Value**: Write **Database-SG**.
 - d. Click **Create security group**.

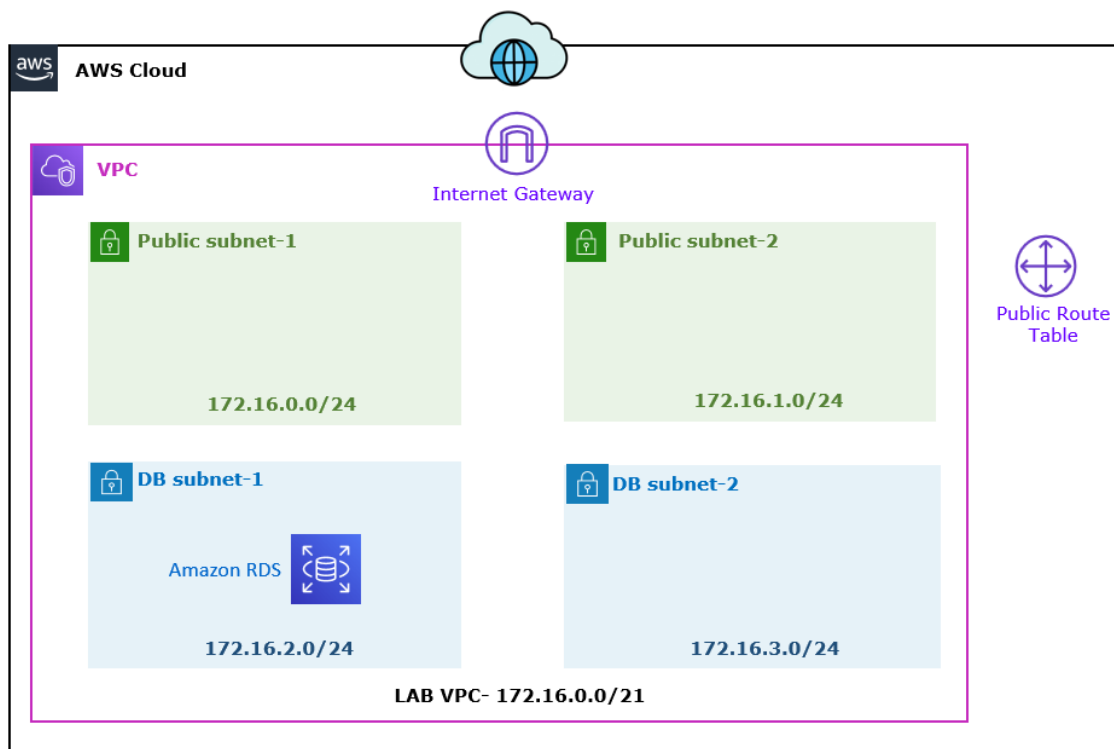
Step 3: Create Secret Manager Security Group

11. **From** the **VPC** console.
12. **Go to the left**, select **Security Groups**.
13. Select **Create security group**.
 - a. In the **Basic details** section:
 - i. **Security group name**: Write **SecretManager-SG**.
 - ii. **Description**: Write **Security Group for Secret Manager**.

- iii. **VPC:** Click on **Cross**.
 - a) Select **LAB VPC**.
- b. In the **Inbound rule** section:
 - ii. Click **Add Rule**.
 - a) **Type:** Dropdown and select **HTTPS**.
 - b) **Source:** Dropdown and select **Anywhere-IPv4**.
- c. In the **Tags - optional** section:
 - iii. Click **Add new tag**:
 - a) **Key:** Write **Name**.
 - b) **Value:** Write **Secretmanager-SG**.
- d. Click **Create security group**.

Task 2: Create Database

In this task you can create db subnet group and mysql database.



Step 1: Create an Amazon RDS Subnet Group

14. In the **AWS Management Console**, on the **Services** menu, click **RDS**.

15. **Go to the left**, select **Subnet groups**.

a. Select **Create DB Subnet Group**.

i. In the **Subnet group details** section:

a) **Name**: Write **DB-Subnet-Group**.

b) **Description**: Write **DB-Subnet Group for MySQL**.

c) **VPC**: Dropdown and select **LAB VPC**.

ii. In the **Add subnets** section:

a) **Availability zone**:

1) Dropdown and select **us-east-1a**.

2) Dropdown and select **us-east-1b**.

b) **Subnets**:

1) Dropdown and select **172.16.2.0/24** CIDR block.

2) Dropdown and select **172.16.3.0/24** CIDR block.

Note: You can see 2 Subnets (172.16.2.0/24 and 172.16.3.0/24) in the subnets selected.

c) Select **Create**.

Step 2: Create an Amazon RDS Instance

16. **Go to the left**, select **Databases**.

17. Select **Create database**.

a. In the **Choose a database creation method** section:

i. Select **Standard Create**.

b. In the **Engine options** section:

i. **Engine type**: Select **MySQL**.

ii. **Version**: Dropdown and **Select 8.0 [Latest version]**.

- c. In the **Templates** section:
 - i. Select **Free tier**.
- d. In the **Settings** section:
 - i. **DB instance identifier**: Write **Inventory-DB**.
 - ii. **Master username**: Write **master**.

Note: This is the **username** you use to login and access your MySQL database instance.

- iii. **Master password**: Write **lab-password**.
- iv. **Confirm password**: Write **lab-password**.

Note: This is the **password** you use to login and access your MySQL database instance.

- e. In the **DB instance class** section:

Note: You can see **db.t2.micro** is selected.

- f. In the **Connectivity** section:
 - i. **Virtual private cloud (VPC)**: Dropdown and select **LAB VPC**.

Note: You can see **db-subnet-group** is selected under **Subnet group**.

- ii. **Public access**: Select **No**.
- iii. **VPC security groups**: Select **Choose existing**.

Note: **Remove** the **default** security group.

- a) **Existing VPC security groups**: Dropdown and select **Database-SG** (which you have created in the previous step).

Note: Ensure only **Database-SG** security group should **display**.

- iv. **Availability zone:** Dropdown and select **us-east-1a**.
- g. In the **Database authentication** section:
 - i. Select **Password authentication**.
- h. In the **Additional configuration** section:
 - i. **Initial database name:** Write **prod_schema**.

Note: Leave other options as default.

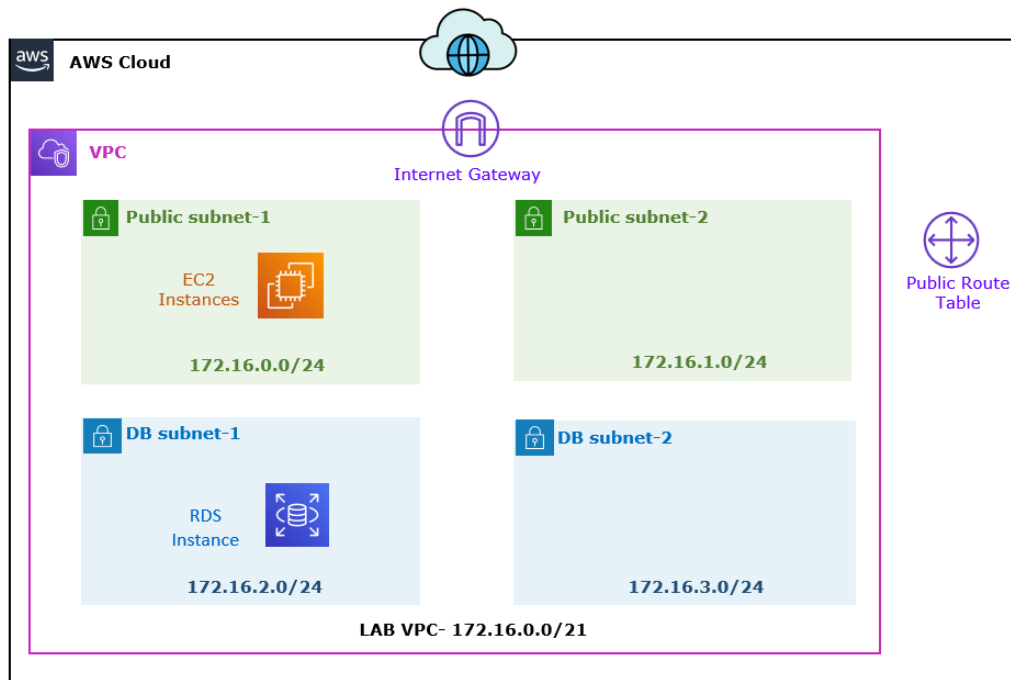
- i. Click **Create database**.

Note: You will see the Database Instance Status as **Creating**.

Note: DB Instance creation will take a **~20 minutes**. **Don't wait, go to the next task.**

Task 3: Create Web Server

In this task you can create web server.



Step 1: Create Ubuntu Instance

18. In the **AWS Management Console**, on the **Services** menu, click **EC2**.

19. Click **Launch Instance**.

a. In the **Choose an Amazon Machine Image (AMI)** page:

- i. Select the **Ubuntu Server 18.04 LTS**.
- ii. Click **Select**.

b. In the **Choose an Instance Type** page:

- i. Select the **t2.micro**.
- ii. Click **Next: Configure Instance Details**.

c. In the **Configure Instance Details** page:

- i. **Network:** Dropdown and select **LAB VPC**.
- ii. **Subnet:** Dropdown and select **Public Subnet-1**.
- iii. **User data:** Select **As file**.
 - a) Click on **Choose file** and **navigate** and select **Script_WebApplication.txt** script file.

Note: **Script_WebApplication.txt** is available with the Lab manual.

Note: Leave the other settings as default.

- iv. Select **Next: Add Storage**.

d. In the **Add Storage** page:

Note: Leave all the settings as default.

- i. Select **Next: Add Tags**.

e. In the **Add Tags** page:

- i. Click on **Add Tag**:

- a) **Key:** Write **Name**.
- b) **Value:** Write **WebApp Server**.
- ii. Select **Next: Configure Security Group**.
- f. In the **Configure Security Group** page:
 - i. **Assign a security group:** Select **Select an existing security group**.
 - a) **Security group name ID:** Select **WebApp-SG**.
 - b) Select **Continue**.
 - ii. Select **Review and Launch**.
- g. In the **Review Instance Launch** page:
 - i. Select **Launch**.
 - a) In the **Select an existing key pair or create a new key pair** section:
 - 1) **First selection box:** Dropdown and select **Choose an existing key pair**.
 - 2) **Key pair name:** Dropdown and select **My-SEC-LAB-KP**.
 - 3) **Enable** the **Checkmark** against the ***I acknowledge that I have access***

Note: Leave the other settings as default.

- ii. Click on **Launch Instances**.
- h. Select the **View Instances**.

Step 2: Check the Web Server Status

20. From the **EC2** console:

21. Click **Instance**.

a. Select **Web Server**.

i. **Wait** for the **Instance State** to change to **Running** state.

ii. **Wait** for the **Status check** to change to **2/2 checks passed**.

Note: **Wait** (~5-10 mnts.) for the server deployment.

b. Select **Web Server**.

i. **Go below** in the console and click on **Details**.

ii. **Expand** the **Instance summary**.

a) **Copy** the **Public IP address** in the **Notepad**.

Task 4: Create Database Schema

In this task you can create database table and schema.

Step 1: Copy the RDS Instance Endpoint

17. In the **AWS Management Console**, on the **Services** menu, click **RDS**.

18. Select **Databases**.

Note: **Wait** till **status** of RDS Instance showing **Available**.

19. Open **Inventory-DB** instance.

a. Select **Connectivity & security**.

b. **Copy** the **Endpoint** under in **Notepad**.

Step 2: Connect to Web Server

20. From the **Local Desktop/ Laptop**, right click on **Start** & **Run**.

a. **Connect** to **Web Server** using the **putty**.

Note: Map your *.ppk file before connecting to **Web Server-01** via **putty**.

- i. **Login as:** Write username as **ubuntu**.

Step 3: Create Table in Database

22. From the **Linux terminal console**:

- a. **Connect** to **RDS MySQL Instance**

`mysql -u master -p -h HOSTNAME`

Note: Replace the **HOSTNAME** with **RDS MySQL instance endpoint**.

- b. You will be **prompted** to **enter** the **password**.

Type **lab-password**

Note: You will be shown a **brief introduction message** and then be placed at the **mysql>** prompt.

23. From the **MySQL console**:

- a. **List** the **existing databases**.

`show databases;`

- b. **Use** the **Inventory database**.

`use prod_schema;`

- c. **Create Products table** in the **Inventory database**.

`create table `products` (
 `id` int(11) not null auto_increment,
 `name` varchar(45) not null,
 `quantity` varchar(45) not null,
 `price` varchar(45) not null,
 primary key (`id`),
 unique key `id_unique` (`id`));`

- d. **List** the tables in the **Inventory database**.

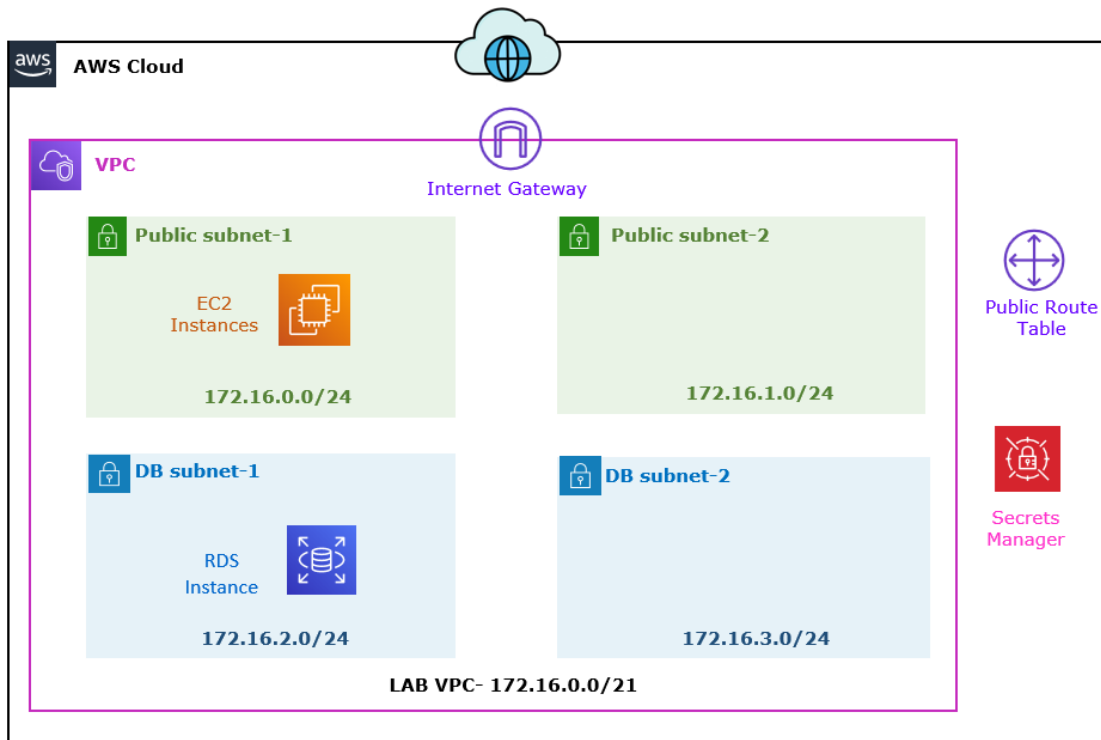
`show tables;`

- e. **Exit mysql instance**.

`exit`

Task 5: Store Database Credentials

In this task you can create secret manager and store database credentials.



Step 1: Create Connection String

24. **Update** the below **Connection String** in **Notepad**.

HOST-NAME,master,lab-password,prod_schema,products

Note: Replace the **HOSTNAME** with **RDS Primary MySQL instance endpoint**.

Step 2: Create Secrets for Storing Database Credentials

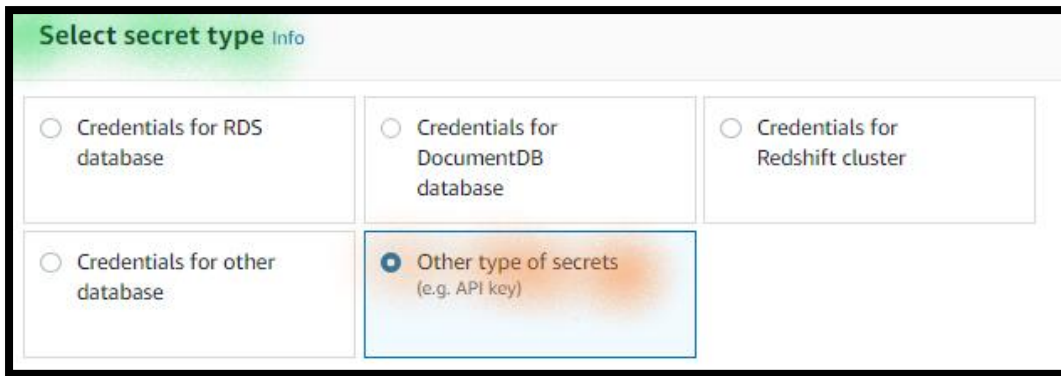
25. In the **AWS Management Console**, on the **Services** menu, click **Secrets Manager**.

26. **Go to the left**, select **Secrets**.

27. Select **Store a new secret** and **Configure**:

a. In the **Secret type** section:

i. Select **Other type of secrets**.



Select secret type Info

☐ Credentials for RDS database

☐ Credentials for DocumentDB database

☐ Credentials for Redshift cluster

☐ Credentials for other database

☒ Other type of secrets (e.g. API key)

b. In the **Key/value pairs** section:

i. Select **Key/value**.

a) **First column:** Write **connection**.

b) **Second column:** Type **CONNECTION STRING** (which you have created in the previous step).

ii. Select **Next**.



Specify the key/value pairs to be stored in this secret Info

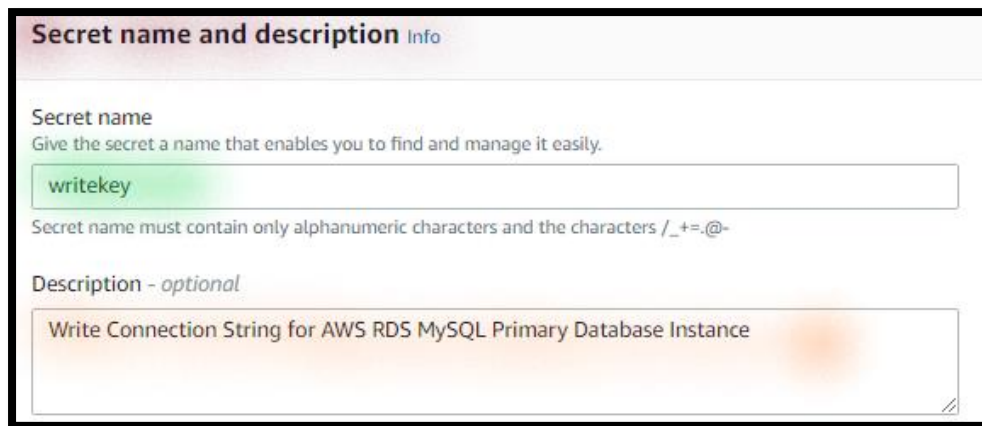
Secret key/value	Plaintext
connection	inventory-db-p.cmc7clgbd1iv.us-east-

+ Add row

c. In the **Secret name and description** section:

i. **Secret name:** Write **rwkey**.

ii. **Description:** Type **Read/ Write Connection String for RDS MySQL Database Instance**.



Secret name and description Info

Secret name
Give the secret a name that enables you to find and manage it easily.

writekey

Secret name must contain only alphanumeric characters and the characters / _ + = . @ -

Description - optional

Write Connection String for AWS RDS MySQL Primary Database Instance

iii. Select **Next**.

d. In the **Secret rotation** section:

Note: Leave all the details as default.

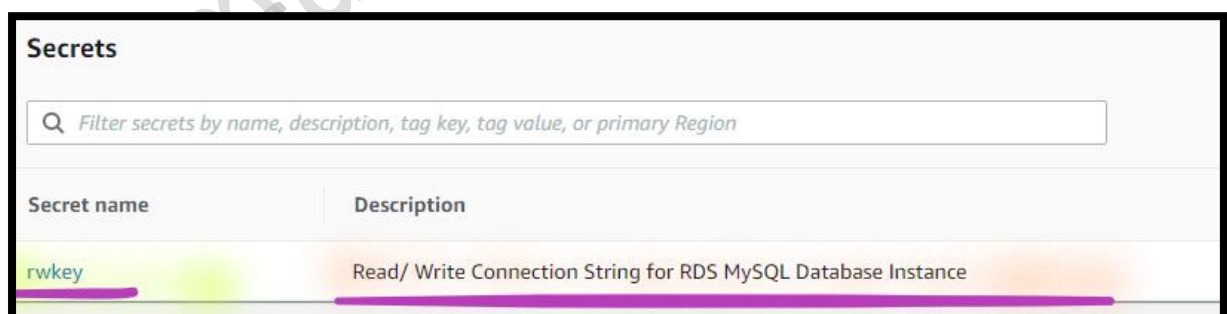
i. Select **Next**.

e. In the **Review** section:

Note: Review the details.

i. Select **Store**.

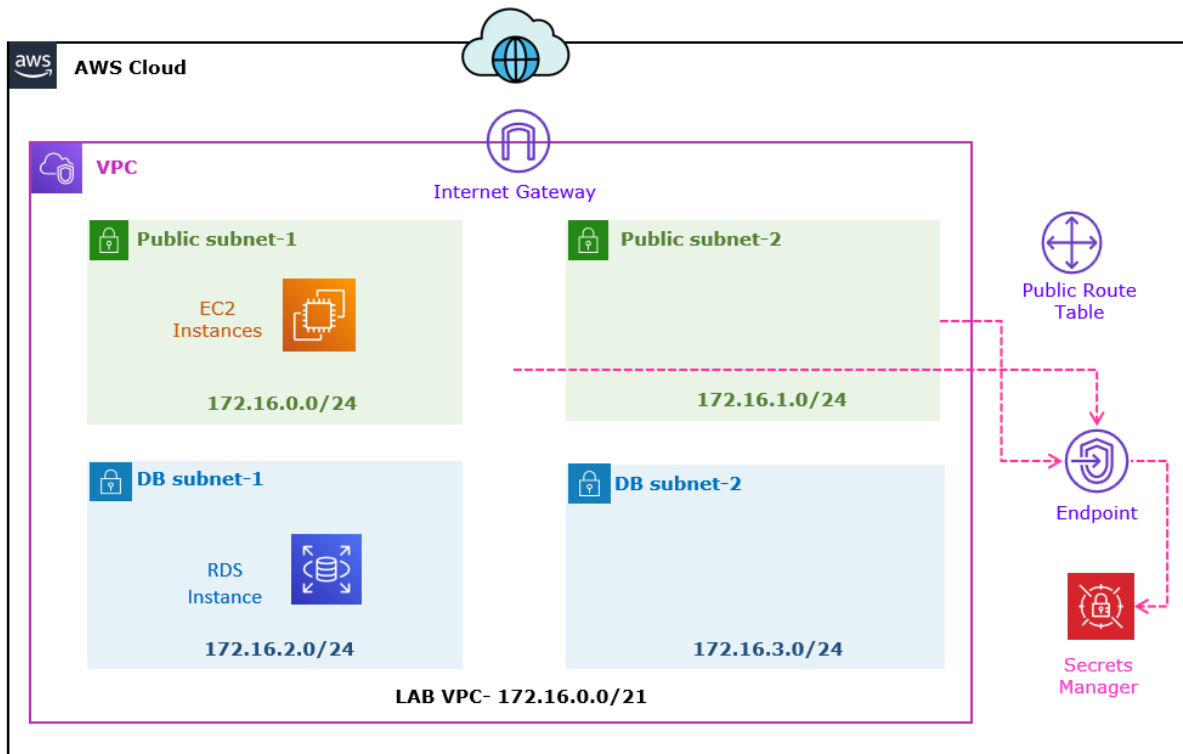
Note: You can see the **secrets**.



Secrets	
Filter secrets by name, description, tag key, tag value, or primary Region	
Secret name	Description
rwkey	Read/ Write Connection String for RDS MySQL Database Instance

Task 6: Create VPC Endpoint

In this task you can create vpc endpoint to access secret manager securely.



Step 1: Create VPC Endpoint for Secret Manager

28. In the **AWS Management Console**, on the **Services** menu, click **VPC**.

29. **Go to the left**, select **Endpoints**.

30. Select **Create Endpoint**.

a. In the **Endpoint settings** section:

i. **Name tag**: Write **SM Endpoint**.

ii. **Service category**: Select **AWS services**.

Name tag - optional
Creates a tag with a key of 'Name' and a value that you specify.

SM Endpoint

Service category
Select the service category

☒ **AWS services**
Services provided by Amazon

☐ **PrivateLink Ready partner services**
Services with an AWS Service Ready designation

☐ **AWS Marketplace services**
Services that you've purchased through AWS Marketplace

☐ **Other endpoint services**
Find services shared with you by service name

b. In the **Services** section:

- i. In the **search box** type **com.amazonaws.us-east-1.secretsmanager** and Press **enter**.

Note: You can see the **com.amazonaws.us-east-1.secretsmanager** after search.

- ii. Select **com.amazonaws.us-east-1.secretsmanager**.

Services (1/1)

Filter services

search: com.amazonaws.us-east-1.secretsmanager X Clear filters

Service Name	Owner	Type
com.amazonaws.us-east-1.secretsman...	amazon	Interface

c. In the **VPC** section:

- i. **VPC:** Dropdown and select **LAB VPC**.

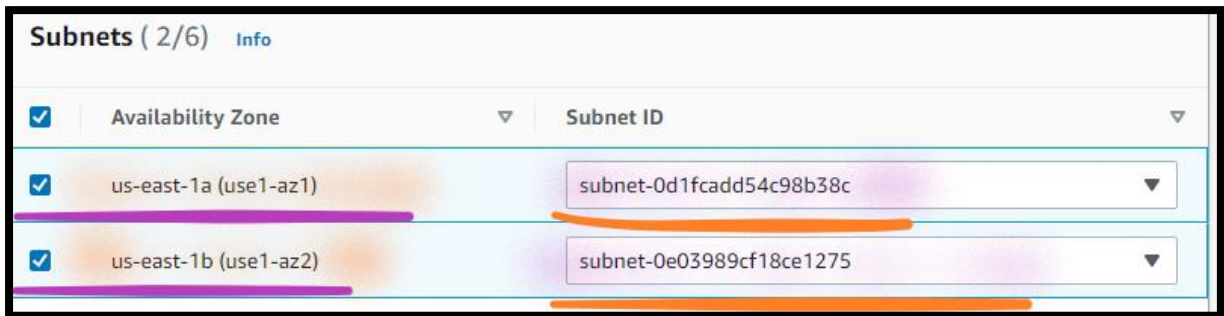
Note: You will now specify which **subnets** the should **privately** access **Secrets manager**.

d. In the **Subnets** section:

- i. Select the **First Availability Zone**:
- a) Dropdown and select **Public subnet-1**.

ii. Select the **Second Availability Zone**:

a) Dropdown and select **Public subnet-2**.

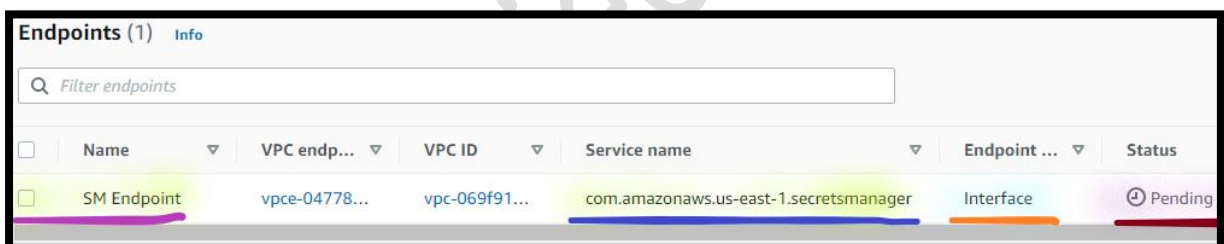


e. In the **Security group** section:

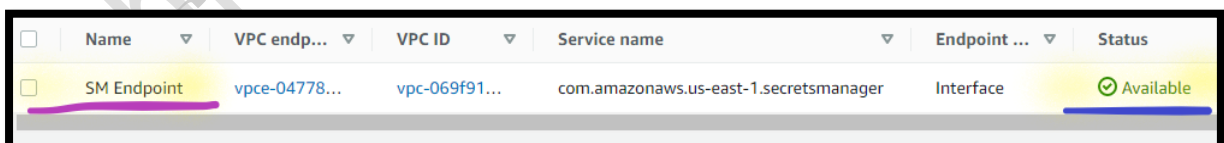
Select the **SecretManager-SG** security group.

f. Select **Create endpoint**.

Note: Service endpoint **Status** shown as **Pending**.

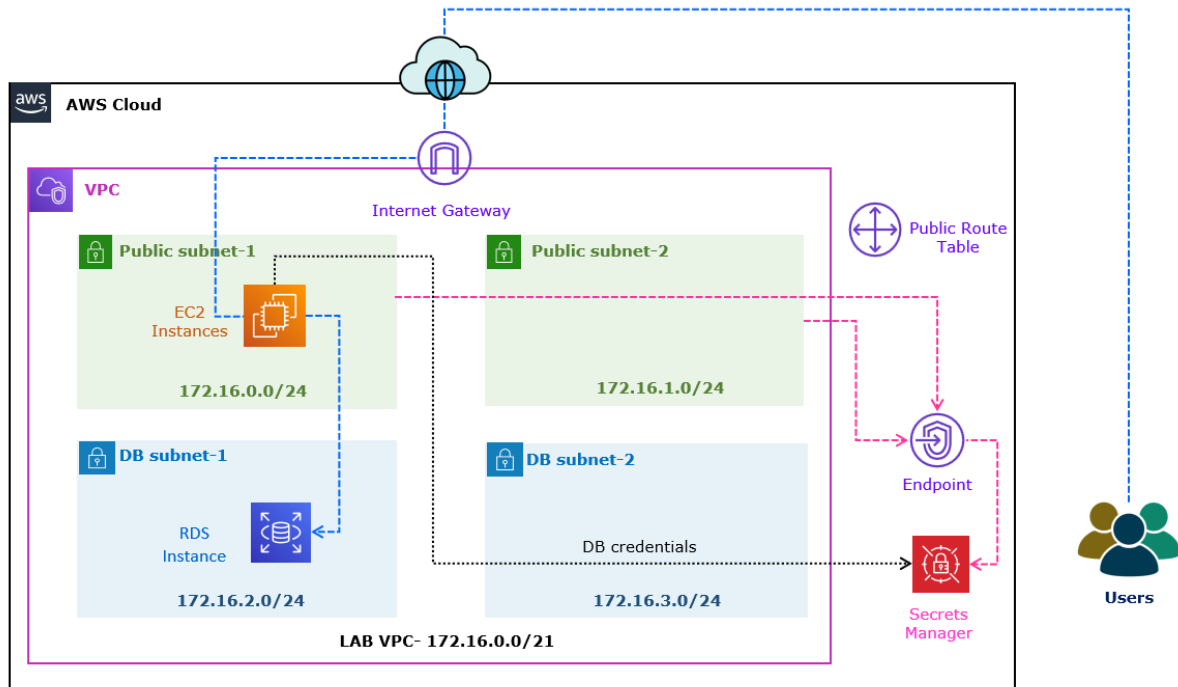


Note: **Wait** (~ 5 mnts.) till, Service endpoint **Status** shown as **Active**.



Task 7: Access the Application

In this task you can access your application. Your web application can connect database securely by fetching the database credentials from secret manager.



Step 1: Create IAM User

Note

- In this task, we are creating IAM user for Amazon Secret manager, to connect database from instances.

31. In the **AWS Management Console**, on the **Services** menu, click **IAM**.

32. Select **Users**.

33. Select **Add user**.

a. In the **Set user details** section:

i. **Username:** Write **Web-SM-User**.

b. In the **Select AWS access type** section:

i. **Select AWS Access type:**

a) Select **Access key - Programmatic access**.

Note: Leave other details as default.

ii. Select **Next: Permissions**.

c. In the **Set permissions** page:

i. Select **Attach existing policies directly**.

a) In the **Search** type **SecretsManagerReadWrite**.

1) Select the **SecretsManagerReadWrite**.

ii. Select **Next: Tags**.

d. In the **Add tags** page:

Note: Leave other details as default.

i. Select **Next: Review**.

e. In the **Review** page:

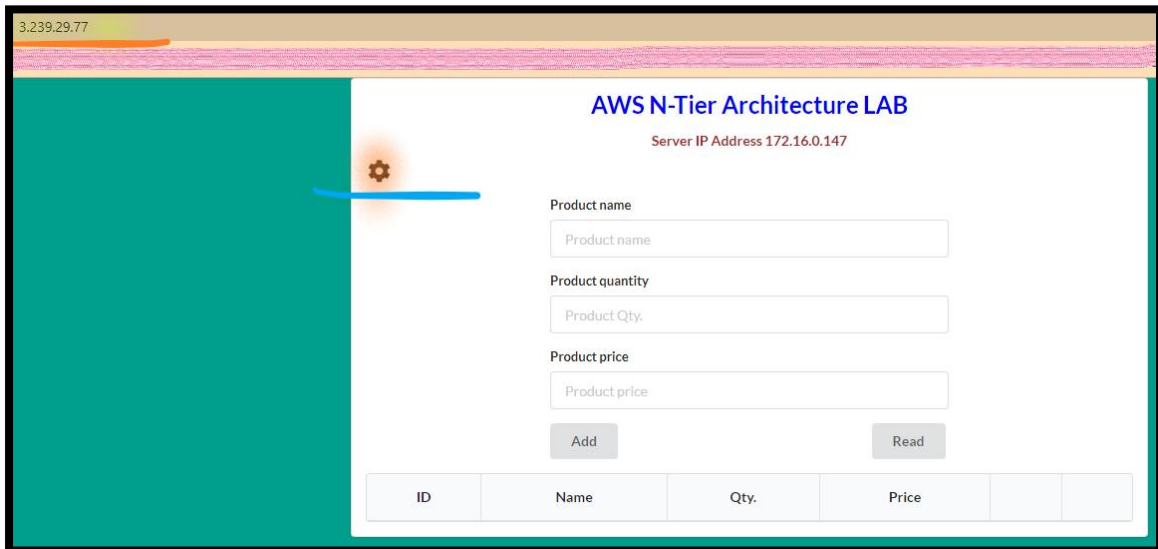
Note: Review all the details.

i. Select **Create users**.

f. Select **Download .csv** file, to download the **Web-SM-User IAM user details**.

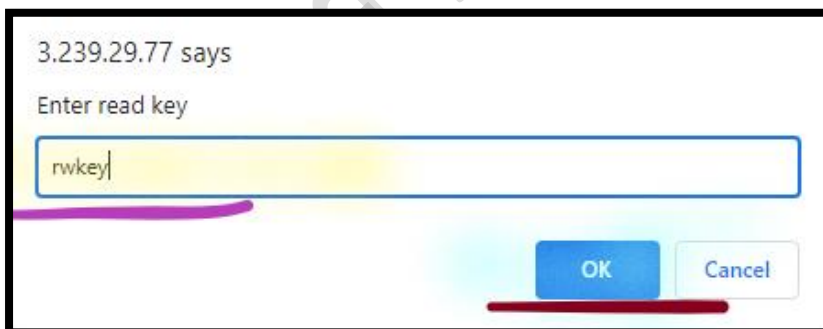
Step 2: Access the Web Server

34. From your **local Desktop/ Laptop Web browser**, type **Public IP address** of the **Web Server**.



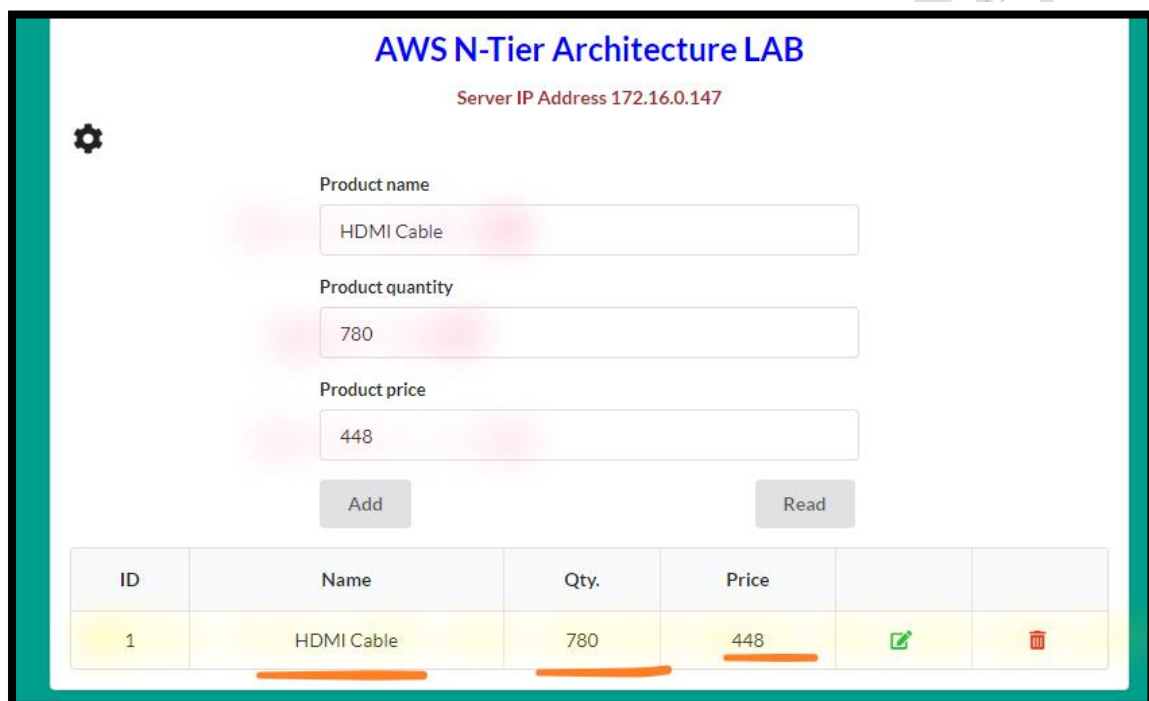
35. From the **Web page**:

- a. Open the **Settings** (**gear icon** in the web page).
 - i. You get the **Prompt** to enter the **Read Key**:
 - a) Write **rwkey**.
 - b) Press **OK**.



- ii. You get the **Prompt** to enter the **Write Key**:
 - a) Write **rwkey**.
 - b) Press **OK**.

- iii. You get the **Prompt** to enter the **Access Key**.
 - a) Write **access key** (copy from the .csv file).
 - b) Press **OK**.
- iv. You get the **Prompt** to enter the **Secret Access Key**.
 - a) Provide **secret access key** (copy from the .csv file).
 - b) Press **OK**.
- b. **Add Product inventory** details.



The screenshot displays the 'AWS N-Tier Architecture LAB' interface. At the top, it shows the 'Server IP Address 172.16.0.147'. Below this, there is a settings gear icon on the left. The main form contains three input fields: 'Product name' with the value 'HDMI Cable', 'Product quantity' with the value '780', and 'Product price' with the value '448'. Below these fields are two buttons: 'Add' and 'Read'. At the bottom, there is a table with the following data:

ID	Name	Qty.	Price		
1	HDMI Cable	780	448		

Task 8: Clean up the Environment

Step 1: Terminate EC2 Instances

36. In the **AWS Management Console**, on the **Services** menu, click **EC2**.
37. Dropdown and select the **US East (N. Virginia)** region.
38. Click **Instances**.
 - a. Select the **Web Server** instances.
 - i. Click on **Instance state**.
 - ii. Select **Terminate instance**.
 - iii. Select **Terminate**.

Step 2: Terminate an RDS

39. In the **AWS Management Console**, on the **Services** menu, click **RDS**.
40. Select the **Databases**.
 - a. Select the **Inventory-DB** database.
 - i. Choose **Actions**.
 - a) Select **Delete**.
 - b) **Unselect** the **Create final snapshot**.
 - c) Select the **I acknowledge**
 - d) To confirm deletion, type **delete me**.
 - e) Choose **Delete**.

Step 3: Delete Secrets

41. In the **AWS Management Console**, on the **Services** menu, click **Secrets Manager**.
42. Open the **rwkey**.
 - a. Select **Actions**.
 - i. Select **Delete secret**.
 - ii. **Waiting period**: Write **7** [**days**].

- iii. Select **Schedule deletion**.

Step 4: Delete Private Endpoint

43. In the **AWS Management Console**, on the **Services** menu, click **VPC**.

44. Open the **Endpoints**.

- a. Select **SM-Endpoint** endpoint.
 - i. Select **Actions**.
 - ii. Select **Delete VPC Endpoint**.
 - a) To confirm deletion, type **delete**.
 - iii. Select **Delete**.

Step 6: Delete Security Groups

45. In the **AWS Management Console**, on the **Services** menu, click **VPC**.

46. Select the **Security groups**.

- a. Select the **WebApp-SG** and **Database-SG** and **SecretManager-SG**.
 - i. Select **Actions**.
 - ii. Select **Delete security groups**.
 - iii. To confirm deletion, type **delete**.
 - iv. Choose **Delete**.

Step 7: Delete the Stack

47. In the **AWS Management Console**, on the **Services** menu, click **CloudFormation**.

- a. Select **Stack**.
 - i. Select **LABVPC**.
 - a) Select **Delete**.
 - b) Select **Delete stack**.